**EX NO. : 10**

**Simulate the network topologies (Bus, Ring, Star and Mesh) using Cisco Packet Tracer**

**Aim :**

To simulate various network topologies (Bus, Ring, Star, Mesh) using Cisco Packet Tracer.

**Procedure:**

**Implementing BUS Topology using Cisco Packet Tracer**

A bus topology is a network in which nodes are directly linked with a common half-duplex link. A host on a bus topology is called a station. In a bus network, every station will accept all network packets, and these packets generated by each station have equal information priority. A bus network includes a single network segment and collision domain.

**Steps to Configure and Setup Bus Topology in Cisco Packet Tracer :**

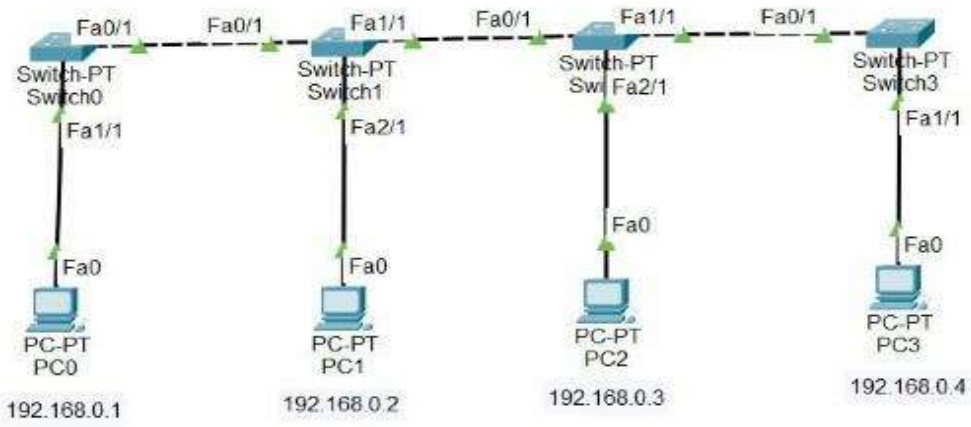
**Step 1:** First, open the Cisco packet tracer desktop and select the devices given below: First, open the Cisco packet tracer desktop and select the devices given below:

| Sl.No | Device | Model-Name |
|---|---|---|
| **1.** | PC | PC |
| **2.** | Switch | PT-Switch |

**IP Addressing Table**

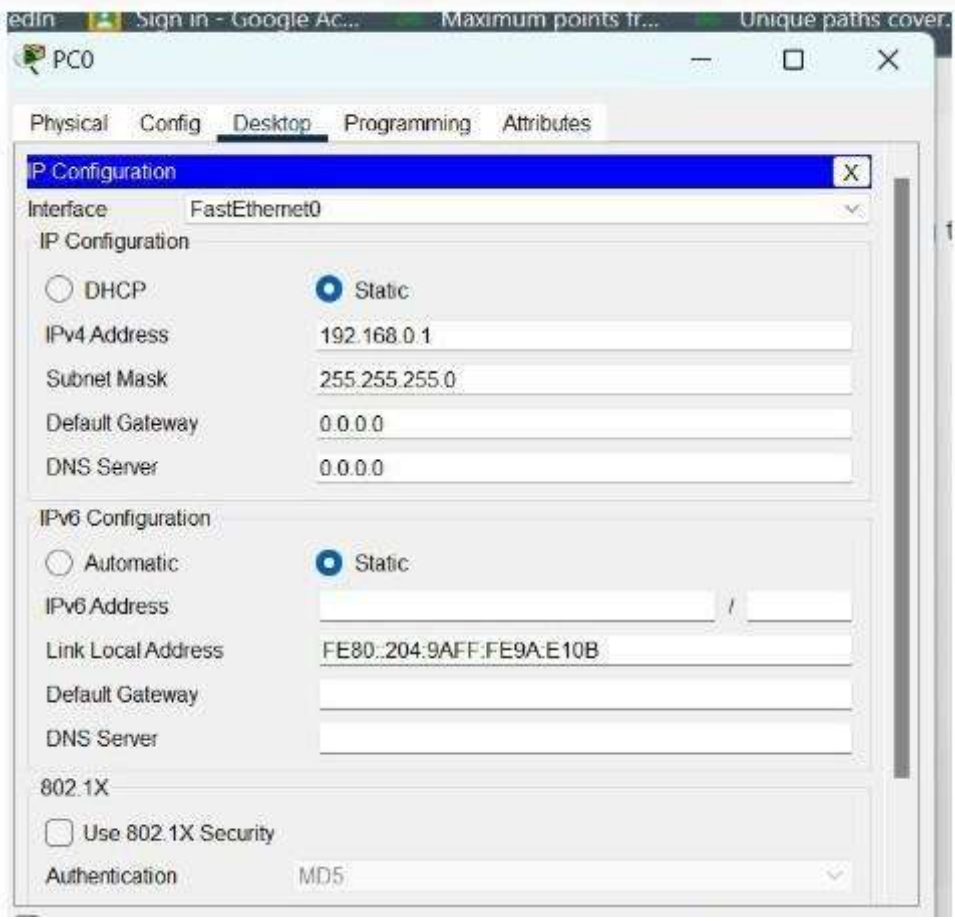
| **S.NO** | **Device** | **IPv4 Address** | **Subnet Mask** |
|---|---|---|---|
| **1** | **pc0** | 192.168.0.1 | 255.255.255.0 |
| **2** | **pc1** | 192.168.0.2 | 255.255.255.0 |
| **3** | **pc2** | 192.168.0.3 | 255.255.255.0 |
| **4** | **pc3** | 192.168.0.4 | 255.255.255.0 |

- Then, create a network topology as shown below image:
- Use an Automatic connecting cable to connect the devices with others.




**Step 2:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

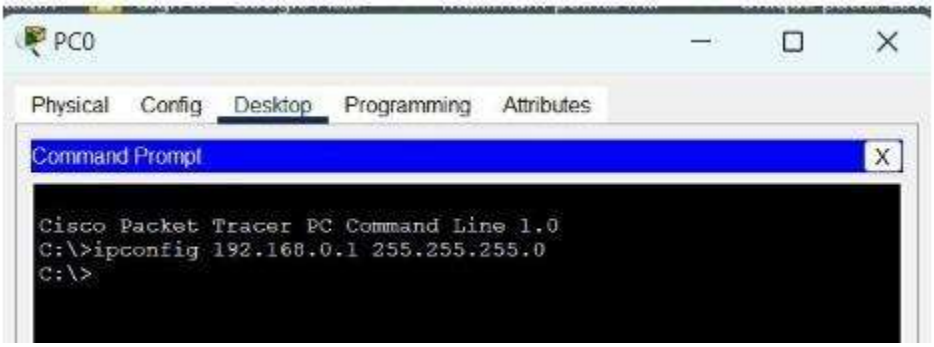
- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.



Assigning an IP address using the ipconfig command, or we can also assign an IP address with the help of a command.

- Go to the command terminal of the PC.
- Then, type ipconfig <IPv4 address><subnet mask><default gateway>(if needed)

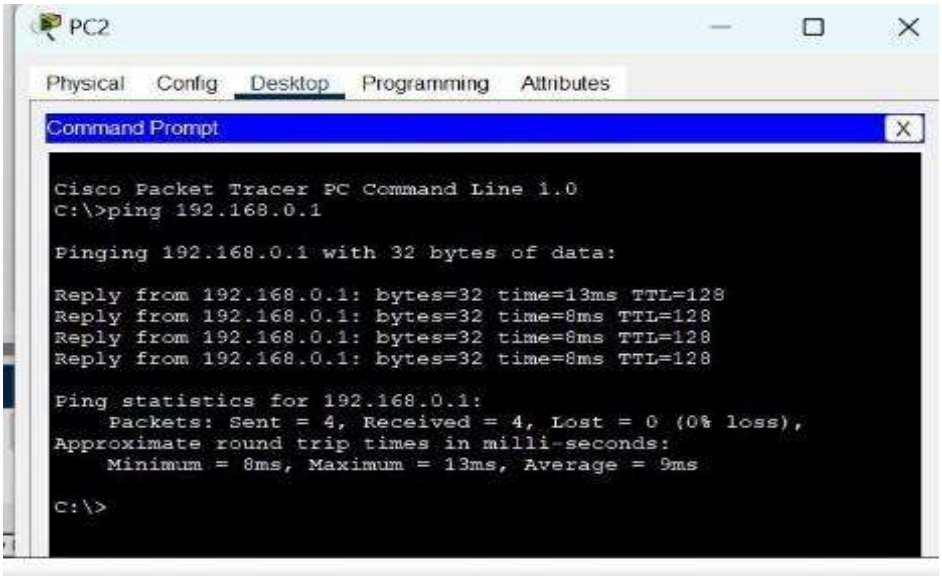
Example: ipconfig 192.168.0.1 subnet mak 255.255.255.0




- Repeat the same procedure with other PCs to configure them thoroughly.

**Step 3:** Verify the connection by pinging the IP address of any host in PC0.

- Use the ping command to verify the connection.




- As we can see we are getting replies from a targeted node on both PCs.

Hence the connection is verified.

**Implementing MESH Topology using Cisco Packet Tracer :**

In the mesh topology of networking, each and every device sends its own signal to the other devices that are present in the arrangement of the network.

**Steps to Configure and Setup Mesh Topology in Cisco Packet Tracer:**

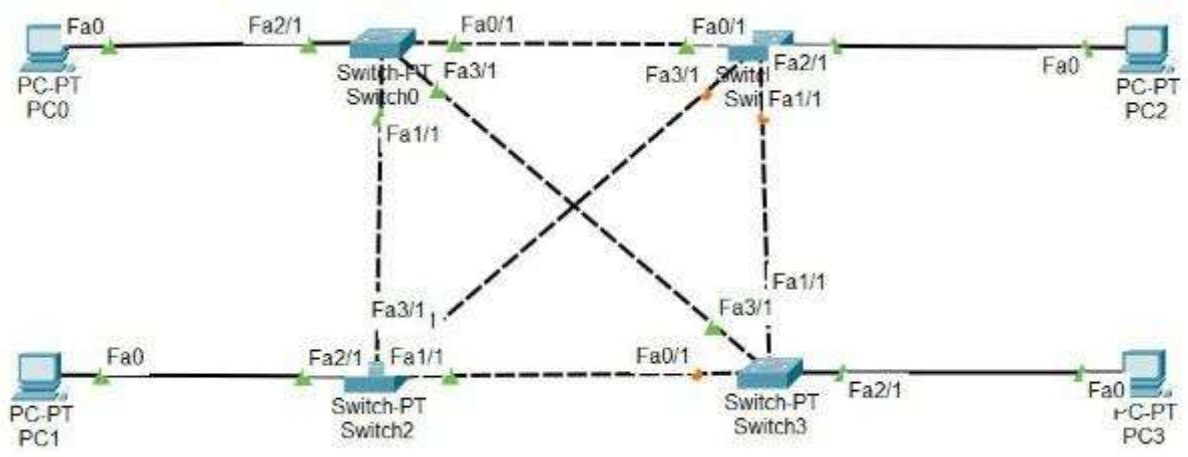
**Step 1:** First, open the Cisco packet tracer desktop and select the devices given below:

| Sl.No | Device | Model-Name |
|---|---|---|
| **1.** | PC | PC |
| **2.** | Switch | PT-Switch |

**IP Addressing Table:**

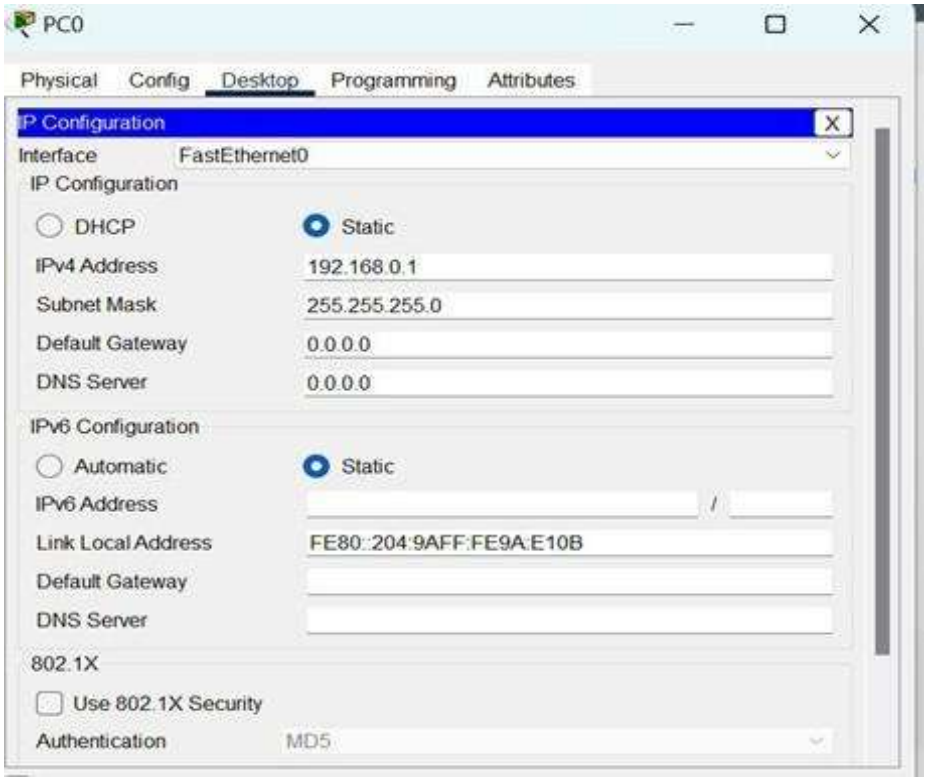
| **S.NO** | **Device** | **IPv4 Address** | **Subnet Mask** |
|---|---|---|---|
| **1** | **pc0** | 192.168.0.1 | 255.255.255.0 |
| **2** | **pc1** | 192.168.0.2 | 255.255.255.0 |
| **3** | **pc2** | 192.168.0.3 | 255.255.255.0 |
| **4** | **pc3** | 192.168.0.4 | 255.255.255.0 |

- Then, create a network topology as shown below the image.
- Use an Automatic connecting cable to connect the devices with others.




**Step 2:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

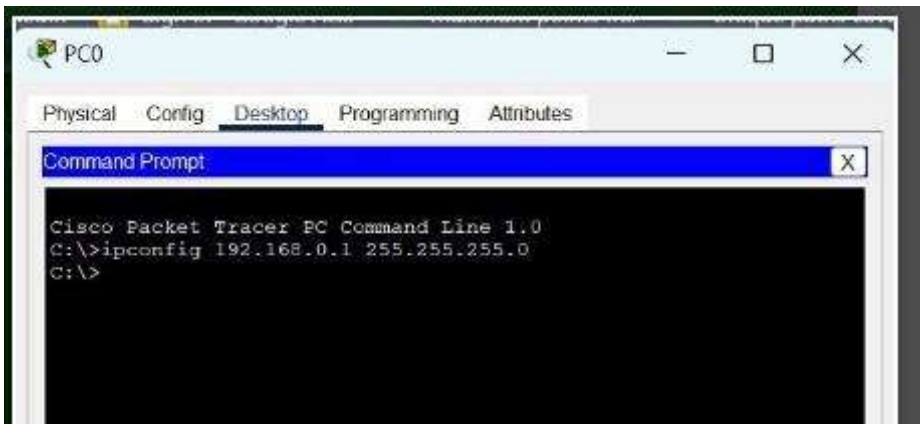
- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.




Assigning IP address using the ipconfig command.

- Also, we can also assign an IP address with the help of a command.
- Go to the command terminal of the PC.
- Then, type ipconfig <IPv4 address><subnet mask><default gateway>(if needed)

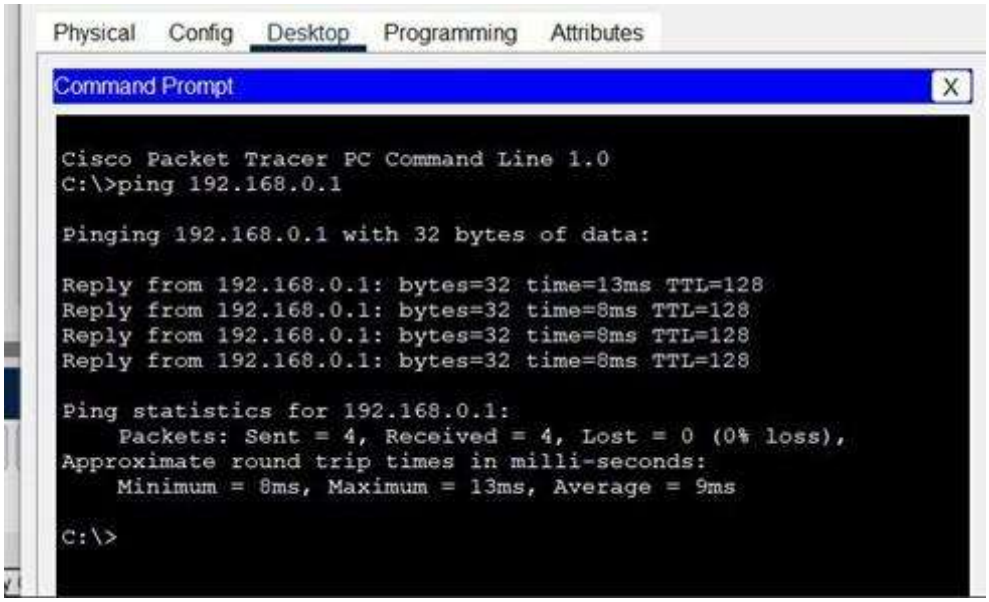
Example: ipconfig 192.168.0.1 255.255.255.0



- Repeat the same procedure with other PCs to configure them thoroughly.

**Step 3:** Verify the connection by pinging the IP address of any host in PC0.

- Use the ping command to verify the connection.
- We will check if we are getting any replies or not.




- Here we get replies from a targeted node on both PCs.
- Hence the connection is verified.

## Implementing Ring Topology using Cisco Packet Tracer

Ring topology is a kind of arrangement of the network in which every device is linked with two other devices. This makes a circular ring of interconnected devices which gives it its name. Data is usually transmitted in one direction along the ring, known as a unidirectional ring. The data is delivered from one device to the next until it reaches the decided destination. In a bidirectional ring, data can travel in either direction.

**Steps to Configure and Setup Ring Topology in Cisco Packet Tracer :**

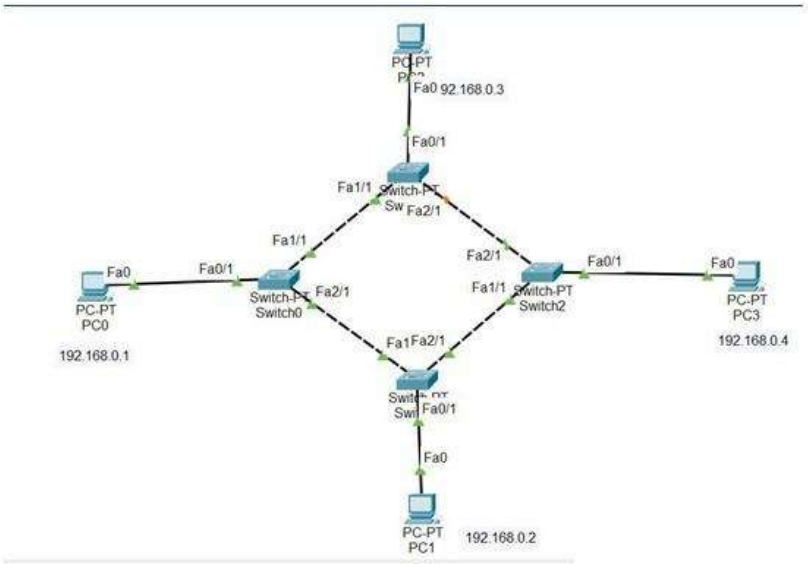
**Step 1:** First, open the cisco packet tracer desktop and select the devices given below:

| Sl.No | Device | Model-Name |
|---|---|---|
| **1.** | PC | PC |
| **2.** | Switch | PT-Switch |

**IP Addressing Table:**

| **S.NO** | **Device** | **IPv4 Address** | **Subnet Mask** |
|---|---|---|---|
| **1** | **pc0** | 192.168.0.1 | 255.255.255.0 |
| **2** | **pc1** | 192.168.0.2 | 255.255.255.0 |
| **3** | **pc2** | 192.168.0.3 | 255.255.255.0 |
| **4** | **pc3** | 192.168.0.4 | 255.255.255.0 |

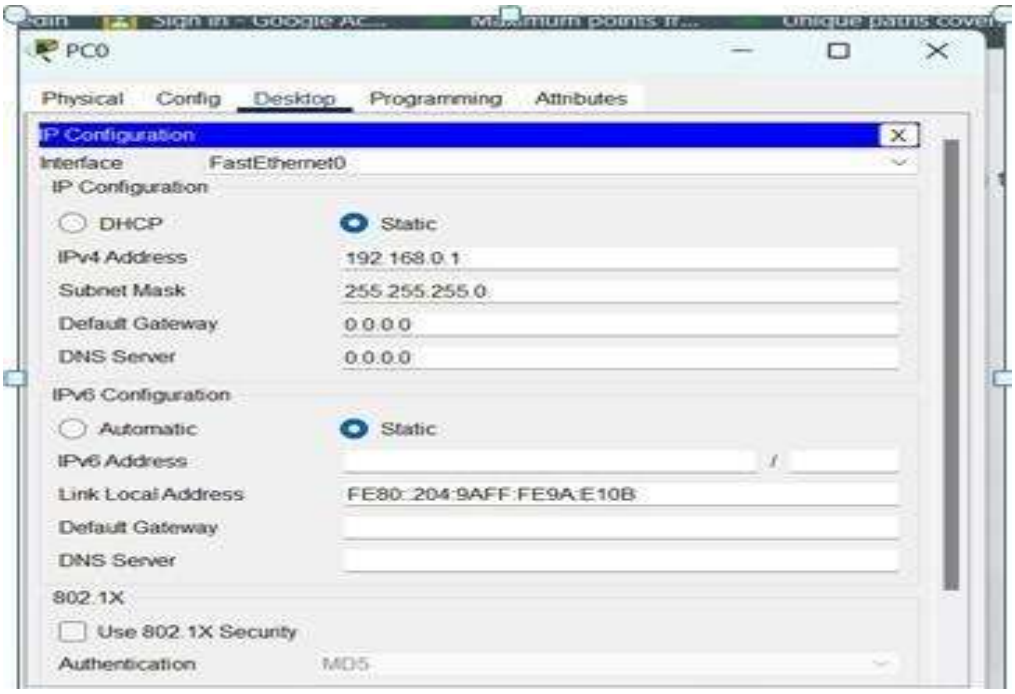
- Then, create a network topology as shown below the image.
- Use an Automatic connecting cable to connect the devices with others.




**Step 2:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

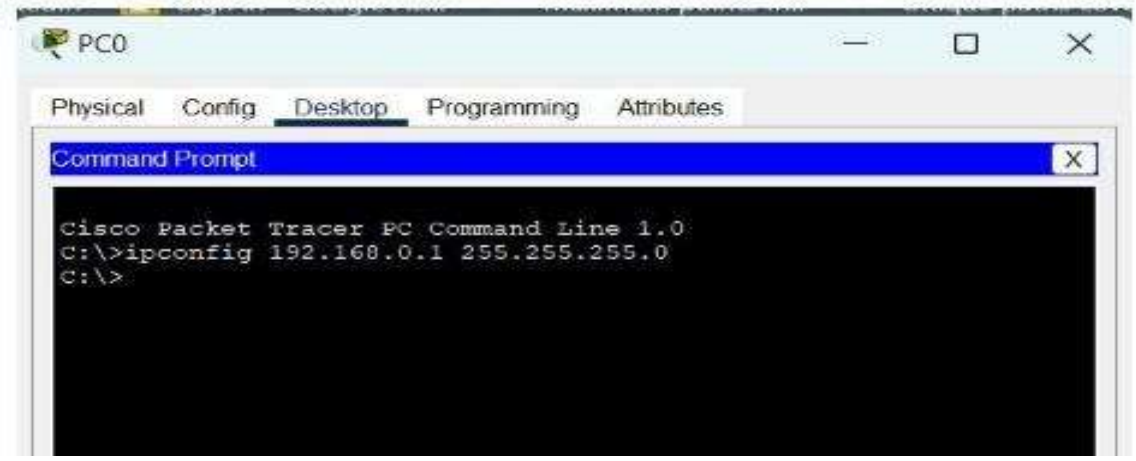
- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.

- Fill IPv4 address and subnet mask.




Assigning IP address using the ipconfig command, or we can also assign an IP address with the help of a command.

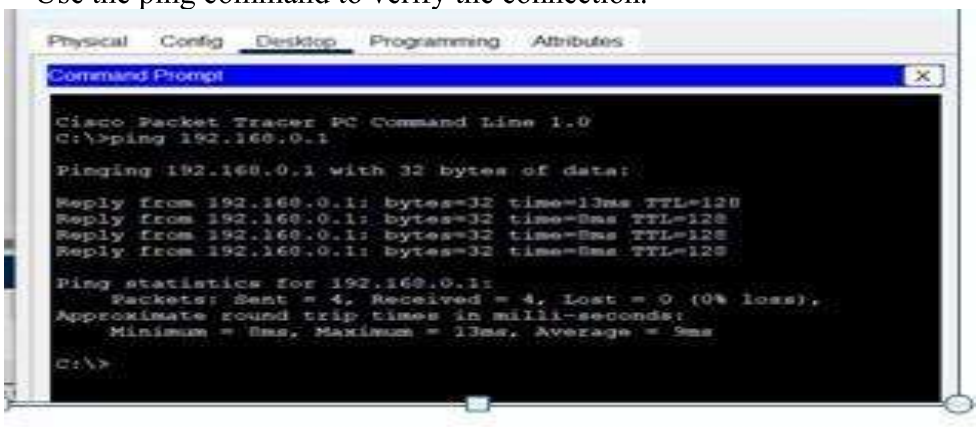
- Go to the command terminal of the PC.
- Then, type ipconfig <IPv4 address><subnet mask><default gateway>(if needed) .
  Example: ipconfig 192.168.0.1 255.255.255.0




- Repeat the same procedure with other PCs to configure them thoroughly.

**Step 3:** Verify the connection by pinging the IP address of any host in PC0.

- Use the ping command to verify the connection.




- As we can see we are getting replies from a targeted node on both PCs.
- Hence the connection is verified.

**Implementing Star Topology using Cisco Packet Tracer**

A star topology for a Local Area Network (LAN) is one in which each node is connected to a central connection point, such as a hub or switch. Whenever a node tries to connect with another node then the transmission of the message must be happening with the help of the central node. The best part of star topology is the addition and removal of the node in the network but too many nodes can cause suffering to the network. A Cisco packet tracer is a simulation tool that is used for understanding the networks. The best part of the Cisco packet tracer is its visualization you can see the actual flow of the message and understand the workflow of the network devices.

**Steps to Configure and Setup Star Topology in Cisco Packet Tracer :**

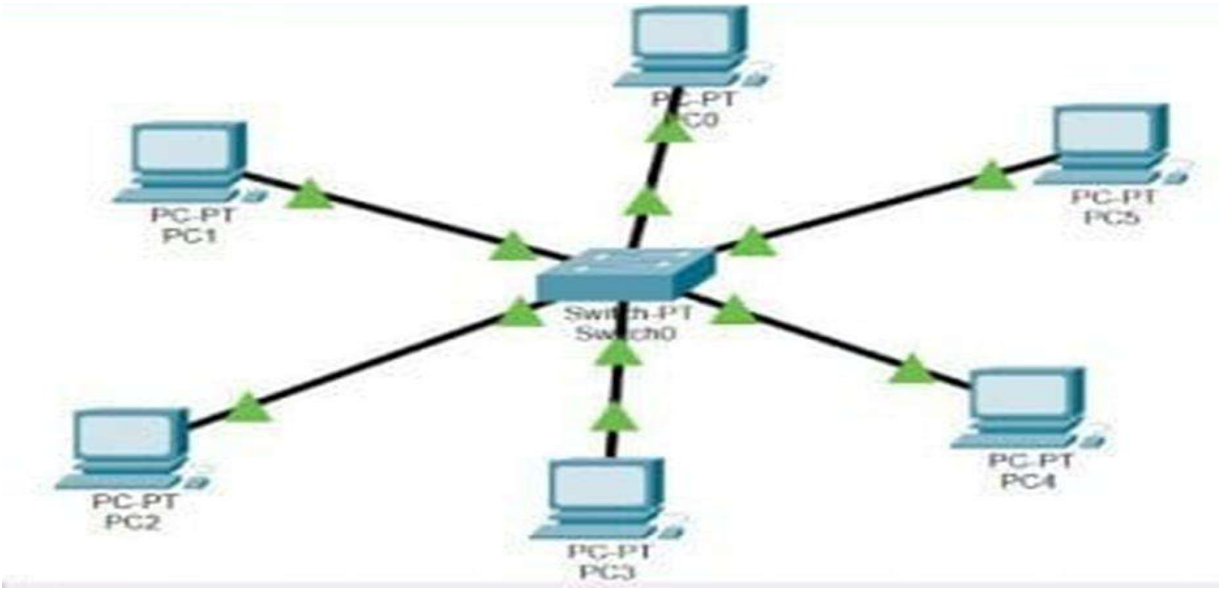
**Step 1:** First, open the cisco packet tracer desktop and select the devices given below:

| Sl.No | Device | Model-Name |
|---|---|---|
| **1.** | PC | PC |
| **2.** | Switch | PT-Switch |

**IP Addressing Table:**

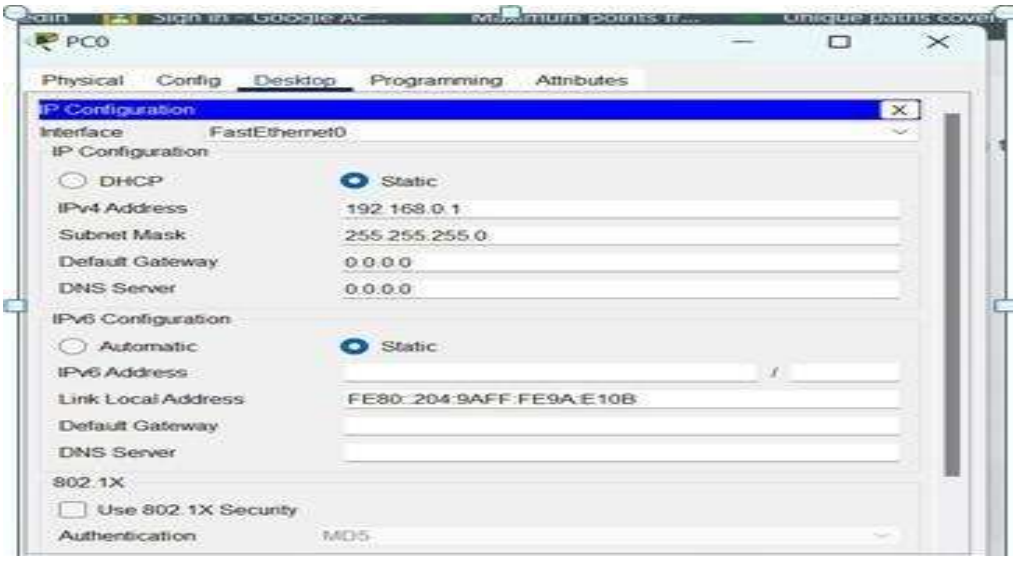
| S.NO | Device | IPv4 Address | Subnet Mask |
|---|---|---|---|
| **1** | **pc0** | 192.168.0.1 | 255.255.255.0 |
| **2** | **pc1** | 192.168.0.2 | 255.255.255.0 |
| **3** | **pc2** | 192.168.0.3 | 255.255.255.0 |
| **4** | **pc3** | 192.168.0.4 | 255.255.255.0 |

- Then, create a network topology as shown below the image.
- Use an Automatic connecting cable to connect the devices with others.



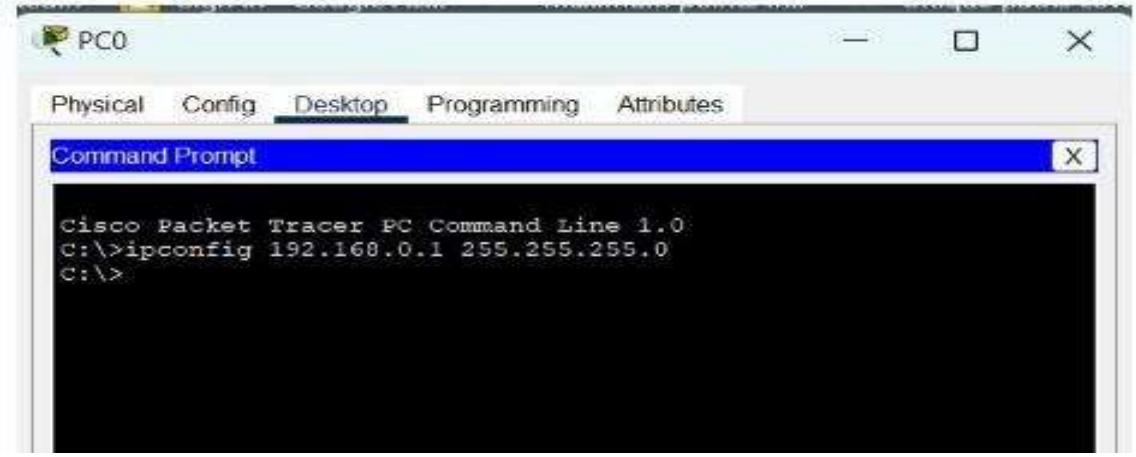

**Step 2:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.




Assigning IP address using the ipconfig command, or we can also assign an IP address with the help of a command.

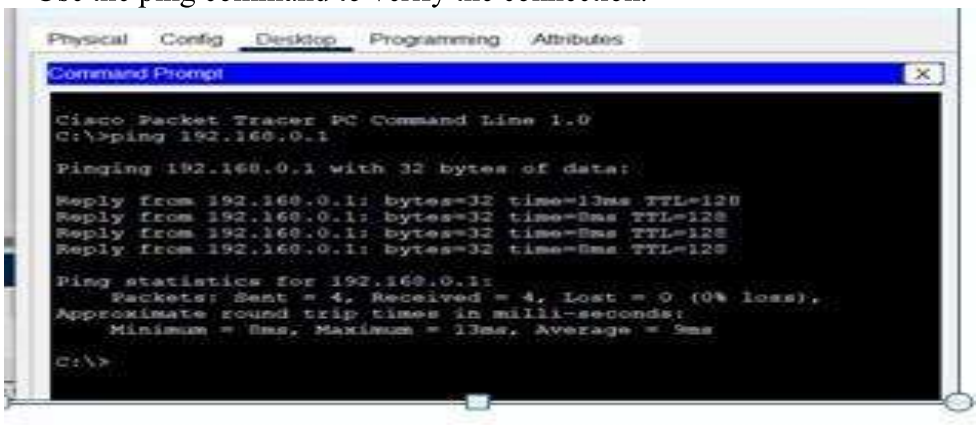
- Go to the command terminal of the PC.
- Then, type ipconfig <IPv4 address><subnet mask><default gateway>(if needed) .
  Example: ipconfig 192.168.0.1 255.255.255.0




Repeat the same procedure with other PCs to configure them thoroughly.

**Step 3:** Verify the connection by pinging the IP address of any host in PC0.

- Use the ping command to verify the connection.




- As we can see we are getting replies from a targeted node on both PCs.
- Hence the connection is verified.

| | |
|---|---|
| Conduct of Experiment (30) | |
| Observation (20) | |
| Viva (10) | |
| Total (60) | |

Result :

Thus the simulation of various network topologies (Bus, Ring, Star, Mesh) using Cisco Packet Tracer has been done.

**EX NO. : 11**

## Implementation of Hub and Switch using Cisco Packet Tracer

**Aim :**

To Simulate and identify the difference in working operation of Hub and Switch using Cisco Packet Tracer.

**Procedure :**

**Implementing Hub using Cisco Packet Tracer :**

A hub is a basic networking device that connects multiple devices in a network. It operates at the physical layer (Layer 1) of the OSI model and works by broadcasting data to all connected devices. Whenever a device sends data, the hub forwards it to every device in the network, regardless of the destination. The best part of a hub is its simple design and low cost, making it easy to set up in small networks. However, due to excessive data transmission, network congestion and collisions may occur, affecting performance.

**Steps to Configure and Setup Hub in Cisco Packet Tracer :**

**Step 1:** First, open the cisco packet tracer desktop and select the devices given below:

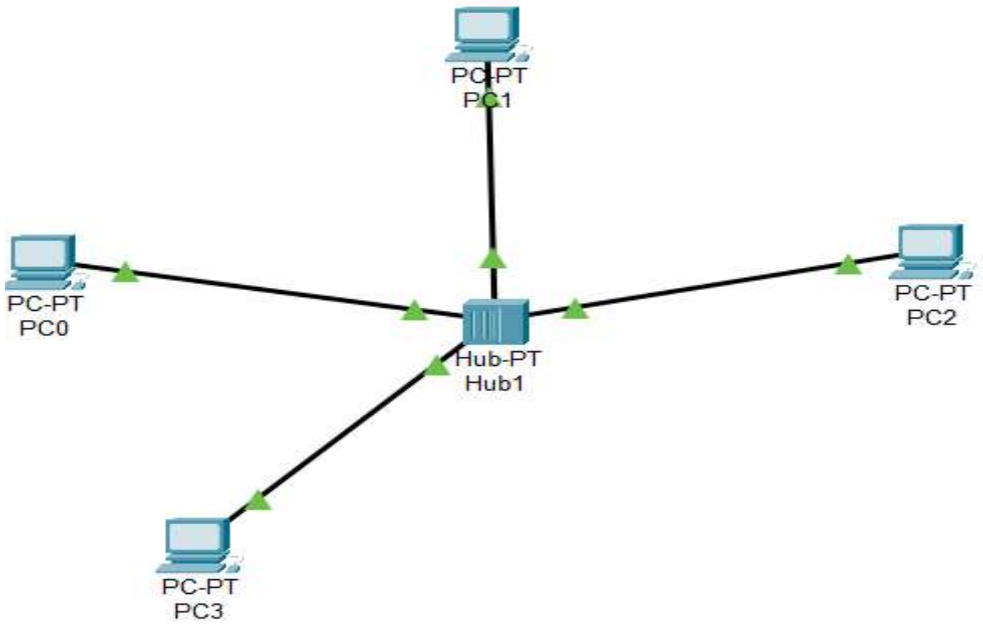
| Sl.No | Device | Model-Name |
|---|---|---|
| **1.** | PC | PC |
| **2.** | Hub | PT-Hub |

**IP Addressing Table:**

| **S.NO** | **Device** | **IPv4 Address** | **Subnet Mask** |
|---|---|---|---|
| **1** | **pc0** | 192.168.0.1 | 255.255.255.0 |
| **2** | **pc1** | 192.168.0.2 | 255.255.255.0 |
| **3** | **pc2** | 192.168.0.3 | 255.255.255.0 |
| **4** | **pc3** | 192.168.0.4 | 255.255.255.0 |

- Then, create a network topology as shown below the image.

- Use an Automatic connecting cable to connect the devices with others.




**Step 2:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.




Repeat the same procedure with other PCs to configure them thoroughly.

Steps to Send a Message from One PC to Another Through a Hub in Cisco Packet Tracer :

Step 1: Open the Simulation Mode

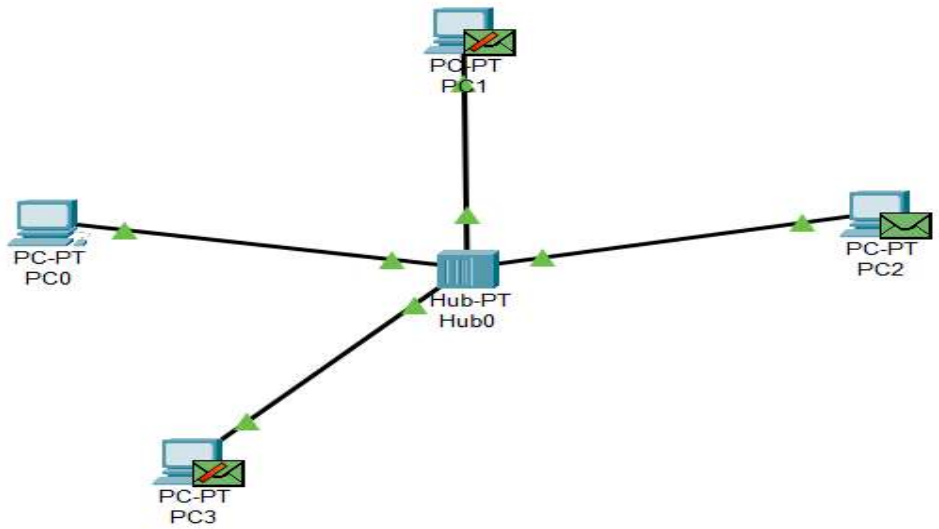
1. Click on the **Simulation Mode** (the clock icon at the bottom-right).
2. This mode allows you to visualize how data is transmitted through the hub.

Step 2: Select the Message Tool

1. Click on the **Add Simple PDU** tool (represented by an envelope icon).
2. Click on the **PC that will send the message** (e.g., PC0).
3. Then, click on the **destination PC** (e.g., PC2).
4. This will create a data packet to be sent.

Step 3: Observe the Data Transmission

1. Click the **Auto Capture / Play** button to see how the packet moves through the network.
2. Since you are using a **hub**, the message will be **broadcast to all connected devices**, not just the intended recipient.
3. The **destination PC** will respond, but all other PCs will also receive the message before ignoring it.



The simulation of message transmission from one PC to another through a **hub** in Cisco Packet Tracer has been successfully implemented. The hub **broadcasts** the message to all connected devices, leading to network congestion. This experiment helps in understanding the **inefficiency of hubs** in handling network traffic.

**Implementing Switch using Cisco Packet Tracer :**

A **switch**, on the other hand, is an **intelligent networking device** that operates at the **data link layer (Layer 2)** of the OSI model. It learns and stores the **MAC addresses** of connected devices, allowing it to forward data only to the intended recipient. Unlike hubs, switches help reduce unnecessary traffic and eliminate collisions, improving network efficiency. The best part of a switch is its **ability to enhance speed and security**, making it ideal for large and modern networks. However, compared to hubs, switches are **more expensive** but offer superior performance.

**Steps to Configure and Setup Hub in Cisco Packet Tracer :**

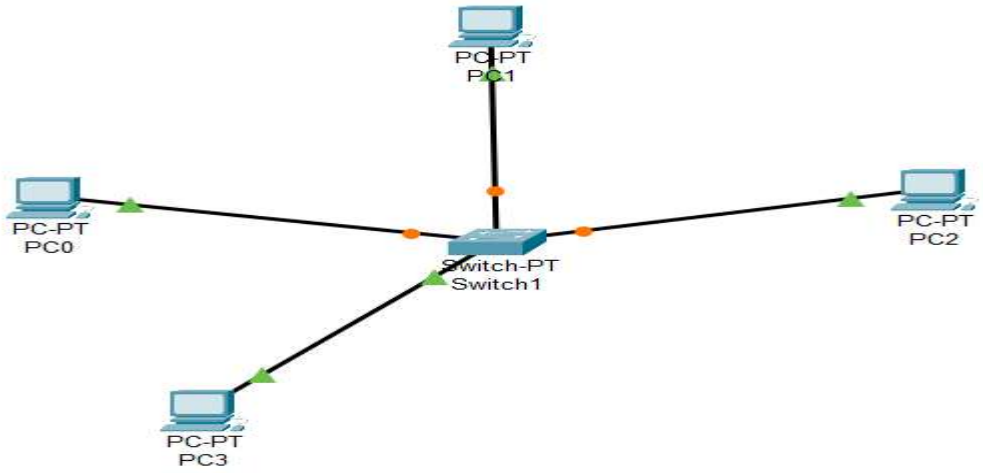
**Step 1:** First, open the cisco packet tracer desktop and select the devices given below:

| Sl.No | Device | Model-Name |
|---|---|---|
| **1.** | PC | PC |
| **2.** | Hub | PT-Switch |

**IP Addressing Table:**

| S.NO | Device | IPv4 Address | Subnet Mask |
|---|---|---|---|
| **1** | **pc0** | 192.168.0.1 | 255.255.255.0 |
| **2** | **pc1** | 192.168.0.2 | 255.255.255.0 |
| **3** | **pc2** | 192.168.0.3 | 255.255.255.0 |
| **4** | **pc3** | 192.168.0.4 | 255.255.255.0 |

- Then, create a network topology as shown below the image.
- Use an Automatic connecting cable to connect the devices with others.




**Step 2:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration




Repeat the same procedure with other PCs to configure them thoroughly.

Steps to Send a Message from One PC to Another Through a Switch in Cisco Packet Tracer

Once you have set up the network using a **switch**, follow these steps to send a message and observe how a switch operates:

Step 1: Open the Simulation Mode

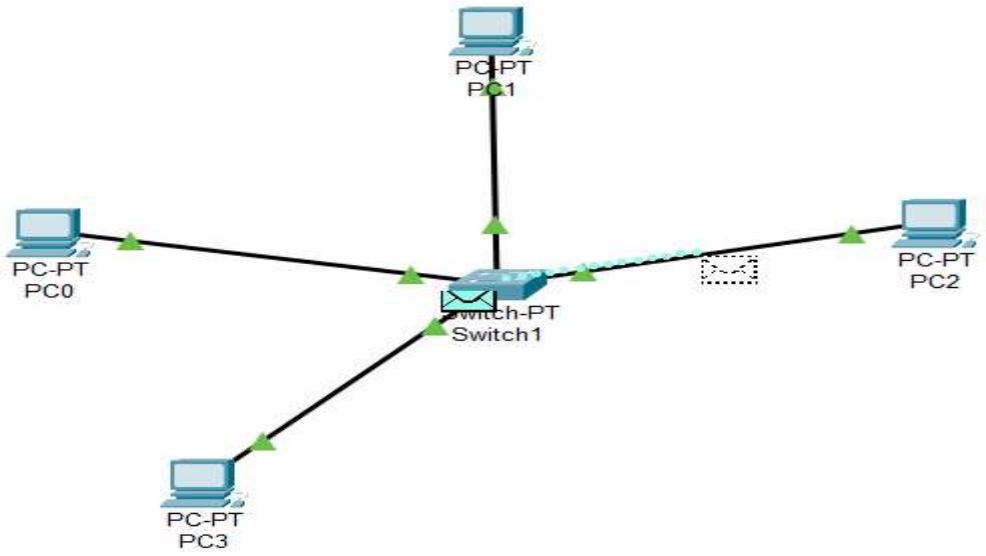
1. Click on the **Simulation Mode** (clock icon at the bottom-right).
2. This mode allows you to visualize how data is transmitted through the switch.

Step 2: Select the Message Tool

1. Click on the **Add Simple PDU** tool (envelope icon).
2. Click on the **PC that will send the message** (e.g., PC0).
3. Then, click on the **destination PC** (e.g., PC2).
4. This creates a data packet to be sent.

Step 3: Observe the Data Transmission

1. Click the **Auto Capture / Play** button to see how the packet moves through the network.
2. Unlike a hub, a **switch intelligently forwards the message only to the intended recipient (**PC2) instead of broadcasting it to all devices.
3. The switch **learns MAC addresses** and ensures efficient data transmission.



The simulation of message transmission from one PC to another through a **switch** in Cisco Packet Tracer has been successfully implemented. The switch **intelligently forwards** the message only to the intended recipient, reducing unnecessary traffic. This experiment helps in understanding the **efficiency of switches** in improving network performance.

| | |
|---|---|
| Conduct of Experiment (30) | |
| Observation (20) | |
| Viva (10) | |
| Total (60) | |

Result :

Thus, the simulation of message transmission has been successfully implemented to **identify the difference in the working operation of a Hub and a Switch** using Cisco Packet Tracer.